WEBSITE BUILD PROMPT

MND CODE

Networking, Cloud & Cybersecurity Academy — MNDCODE.in

Complete, reusable prompt describing the website as built: brand, design system, every page, course curricula, interactive lab, student and admin dashboards, mobile behaviour and acceptance checks. Paste it into an AI builder or hand it to a developer.

1. Role and objective

Act as a senior full-stack developer, UI/UX designer, creative technologist, cybersecurity educator, SEO specialist and software architect. Design and build a complete, production-oriented educational website for **MND CODE**, specialising in networking, cloud computing, cybersecurity and ethical hacking.

Build a connected multi-page learning platform, not a landing page. Every button, card, menu item and CTA must do something real. The result must look clearly better than a typical training-institute website and work equally well on phones, tablets and desktops.

2. Brand identity

Item	Value
Brand name	MND CODE
Company	MND CODE Pvt Ltd
Domain	MNDCODE.in (canonical https://mndcode.in)
Established	01 April 2023 (as supplied by the client)
Tagline	Connect. Compute. Secure.
Supporting headline	Build Your Future in Networking, Cloud & Cybersecurity.
Footer line	© MND CODE — All Rights Reserved
Positioning	A modern technical education platform helping students and aspiring IT professionals build practical skills in computer networking, cloud infrastructure, cybersecurity and authorized penetration testing.

Logo: a rounded-square mark with a gradient outline (blue → cyan → purple) containing a stylised “M” drawn as a network path with three glowing node dots. Wordmark “MND CODE” in Space Grotesk bold, with the small mono sub-line “CONNECT · COMPUTE · SECURE” (hidden on phones).

3. Design system

Aesthetic

Futuristic, premium “network operations” academy: dark editorial layouts, subtle glassmorphism, bento grids, spatial depth and restrained cyber-inspired effects. One light “paper” band on the homepage adds contrast and depth. Avoid excessive glow, oversized text, clutter and animation that interferes with reading.

Colour tokens

Token	Hex	Use
Background	#070A12	Page background
Surface	#101625 / #151D31 / #1B2540	Cards, panels, toasts
Primary blue	#3B82F6	Primary buttons, links, borders
Electric cyan	#22D3EE	Accents, active states, eyebrows, focus ring
Purple accent	#8B5CF6	Security themes, secondary glows
Main text	#F8FAFC	Headings, body on dark
Secondary text	#94A3B8 (dim #64748B)	Descriptions, captions
Success / warning / danger	#22C55E / #F59E0B / #F43F5E	States, chips, alerts
Borders	rgba(148,163,184,.13–.24)	Translucent blue-grey hairlines
Paper band	#EEF2F8 with ink #0B1220	Light learning-process section

Typography

- Headings: **Space Grotesk** 500–700, tight letter-spacing (–0.015em), balanced wrapping.
- Body: **Inter** 400–700, 16 px base, line-height 1.6, running text max ~65–72 characters.
- Utility / data: **JetBrains Mono** for eyebrows, labels, IP addresses, code and telemetry; uppercase labels get 0.08–0.14em tracking.
- Scale: 12 / 14 / 16 / 18 / 22 / 30 px, section headings clamp(2rem, 4vw, 2.75rem), hero clamp(2.3rem, 5.6vw, 4.1rem).
- Load from Google Fonts with system fallbacks.

Visual effects and motion

- Animated gradient lighting: soft blurred “glow orbs” drifting slowly behind page heads.
- Fixed background: subtle 56 px grid with radial blue and purple light, fading out down the page.
- Hero: canvas animation of interconnected network nodes, routers, servers, cloud regions and a security shield exchanging glowing packets, with a “SIMULATED TELEMETRY” tag and three readouts (latency, link health, policy: default-deny).
- Custom SVG illustrations per course (topology map, multi-AZ VPC, shield with scan line).
- Animated dashed connection lines on the learning roadmap; scroll-triggered slide-in reveals (content always visible at rest).
- Short page-transition animation; hover lift on cards; glass header with backdrop blur.
- Respect **prefers-reduced-motion**: disable all animation and smooth scrolling.

Components

Buttons (primary gradient, ghost, quiet, danger; sm/md/lg), cards and card-links, chips (level, category, status), demo flags (dashed amber “Sample project”, “Demo mode” labels), icon boxes, accordions, tabs, progress bars, stat tiles, alerts (ok/bad/warn/info), callouts, tables, modal dialogs, toasts, search palette, breadcrumbs, empty states, spinners. Icons: Lucide-style 2 px strokes.

4. Technology and delivery

- **Prototype deliverable:** one self-contained **index.html** with embedded CSS and JavaScript (no build step), hash routing (e.g. *index.html#/courses/networking*) so it runs on any static host.

- Single data source: one course data structure powers course cards, detail pages, search, lesson player and dashboards.
- Prototype persistence: browser localStorage with in-memory fallback, clearly labelled as demo. Passwords stored only as salted SHA-256 hashes, never plain text.
- Seeded demo accounts shown on the sign-in page: student@demo.mndcode.in / Student#2026 and admin@demo.mndcode.in / Admin#2026 (remove before launch).
- **Production target:** React + TypeScript + Vite (or Next.js), Tailwind CSS, Framer Motion, Lucide icons; Node.js with NestJS; PostgreSQL; REST API with validation; S3-compatible storage for course assets. No secrets in front-end code.
- Site settings object (SITE) holds company info, contact details (null = hidden), social links (only real URLs) and the announcement.

5. Site map and routes

Area	Routes
Public	/ · /about · /courses · /courses/networking · /courses/cloud-computing · /courses/ethical-hacking · /learning-paths · /labs (?t=exercise) · /projects · /resources (?r=item) · /blog (?page=n) · /blog/:slug · /faq · /contact · /privacy · /terms
Student	/login · /register · /dashboard · /dashboard/courses · /learn/:course/:lesson · /dashboard/progress · /dashboard/settings
Admin	/admin · /admin/courses · /admin/lessons (?c=course) · /admin/students · /admin/messages (?f=unread|read)
System	404 page (“destination host unreachable”), error page with retry, sign-in-required page, admin-access-denied page

Student and admin routes are guarded in the prototype and must be enforced on the server in production (front-end hiding is not security). Lessons can be previewed while signed out; tracking requires sign-in.

6. Global layout

- **Announcement bar** (configurable, dismissible, editable in admin): label, text, internal link. No fabricated deadlines or promotions.
- **Sticky glass header:** logo, Home, Courses dropdown (three courses with level, module count and estimated hours, plus “All courses”), Learning Paths, Labs, Resources, About, Contact; search button (Ctrl/■ K); Sign In or the user's name. Active link has a gradient underline.
- **Mobile menu** below 1080 px: hamburger toggles a full-screen list with all pages and course sub-links; body scroll locked while open.
- **Search palette** (modal): searches courses, modules, every lesson, resources, blog posts, projects, lab exercises and key pages; arrow-key navigation, Enter to open, helpful empty state.
- **Footer:** logo, short description, company · domain · established date; columns for Courses, Learn, Company, Contact (shows “Verified contact details will appear here” until configured); social links only when real URLs exist; © MND CODE — All Rights Reserved.
- Breadcrumbs on inner pages, skip-to-content link, back-to-top button, toasts, accessible confirmation dialogs (never browser alert/confirm).

7. Homepage (in this order)

Section	Content
A. Announcement	Configurable bar, e.g. “The interactive lab now includes 10 browser-based exercises — subnetting to report writing. Open the lab →”
B. Navigation	As in section 6.
C. Hero	Eyebrow “TECHNICAL SKILLS FOR THE NEXT GENERATION”. H1 “Master Networks. Build in the Cloud. Secure the Future.” (first line gradient text). Description: “Learn how modern networks operate, how cloud infrastructure connects the world, and how security professionals identify and reduce vulnerabilities through practical, responsible training.” Buttons: Explore Our Courses, Explore Practical Labs. Meta line: 3 structured courses · module count · 10 browser lab exercises. Right side: animated network canvas.
D. Highlights	8 feature cards: Structured learning paths, Hands-on exercises, Realistic guided projects, Beginner-friendly explanations, Progress tracking, Responsible security practice, Command-line confidence, Clear about what's included. No ratings or learner counts.
E. Featured courses	Three large cards: illustration, category chip, level, title, summary, modules / lessons / estimated hours (calculated from data), 3 outcomes, View course.
F. Roadmap	IT Fundamentals → Networking → Cloud Infrastructure → Security Fundamentals → Practical Projects (animated connectors), link to Learning Paths.
G. Labs preview	The real subnet calculator embedded live, plus 4 cards linking to OSI explorer, packet journey, DNS simulator and Linux quiz.
H. Projects	3 example project cards labelled “Educational examples / Sample project”.
I. Why MND CODE	Bento grid: the nine-part lesson structure (large tile with numbered chips), Practice-first, Responsible by design, Platform-neutral cloud, Transparent curriculum.
J. Learning process	Light paper band: Discover → Learn → Practice → Build → Review, each with one sentence.
K. FAQ preview	6 accordion questions: prerequisites, order, equipment, are labs real, certification, legality.
L. Final CTA	“Your Next Chapter in Technology Starts Here.” Buttons: Browse Courses, Contact Us.
M. Footer	As in section 6.

8. Course catalogue and curricula

Each course object holds: slug, title, category, level, short description, overview, audience, prerequisites, learning objectives, outcomes, skills, tools/software, course FAQs, related courses, next course and modules. Each module has a title, summary, lesson topics, 4 key terms and one check question (prompt, options, answer, explanation); project and optional modules are flagged. Lesson counts and hours are always calculated (≈35 min per lesson, 90 min for project lessons) and labelled as estimates.

Course 1 — Network Setup & Administration (/courses/networking)

Category Networking · Beginner · 10 modules · 67 lessons · ≈48 h. Audience: beginners, IT support aspirants, networking students, aspiring network administrators. Next: Cloud.

- **Module 1 — Networking Fundamentals:** What is a computer network?; LAN, WAN, WLAN, PAN and MAN; topologies; client-server and peer-to-peer; network devices; Ethernet, Wi-Fi and media; bandwidth, throughput, latency, jitter, packet loss.
- **Module 2 — OSI Model & TCP/IP:** Seven OSI layers; TCP/IP architecture; encapsulation and decapsulation; TCP vs UDP; HTTP/HTTPS, DNS, DHCP, SSH, SMTP, ICMP; how packets travel; troubleshooting by layer.
- **Module 3 — IP Addressing & Subnetting:** IPv4 and IPv6; public and private addressing; binary-to-decimal; CIDR; subnet masks and calculations; network, host and broadcast addresses; default gateways; IPv6

prefixes; practical exercises.

- **Module 4 — Routing & Switching:** L2/L3 concepts; MAC and ARP; VLANs and trunking; inter-VLAN routing; static routing; intro to dynamic routing; routing tables; segmentation.
- **Module 5 — Network Services:** DNS resolution; DHCP scopes and leases; NAT and PAT; NTP; directory concepts; service troubleshooting.
- **Module 6 — Wireless Networking:** Wi-Fi standards; APs and controllers; SSIDs and channel planning; WPA2/WPA3; guest networks; coverage and interference.
- **Module 7 — Network Security:** Segmentation; firewalls; ACLs; VPNs; SSH administration; least privilege; secure configuration; monitoring and logging.
- **Module 8 — Troubleshooting & Monitoring:** ping, traceroute, ipconfig/ifconfig, ip, ss, nslookup/dig; routing tables; Wireshark basics; methodology; DNS/gateway/addressing problems; documenting incidents.
- **Module 9 — Network Design Projects:** Small-office design; IP plan; segmented topology; simulated VLAN and routing; diagram and troubleshooting report.
- **Module 10 — Assessment & Revision:** Knowledge checks; configuration tasks; troubleshooting scenarios; final design project; review checklist.

Course 2 — Cloud Computing & Networking (/courses/cloud-computing)

Category Cloud · Intermediate · 10 modules · 71 lessons · ≈47 h. Platform-neutral first, then optional AWS, Azure and Google Cloud examples; paid services flagged with guidance on avoiding charges. Next: Ethical Hacking.

- **Module 1 — Cloud Computing Fundamentals:** What cloud is; characteristics; IaaS/PaaS/SaaS; public, private, hybrid, multi-cloud; regions, AZs, edge; shared responsibility; costs.
- **Module 2 — Cloud Platforms & Core Services:** AWS, Azure, Google Cloud; compute, storage, networking, identity; consoles and CLIs; resource organisation; deployment; choosing services.
- **Module 3 — Virtual Networks & Addressing:** VPCs/VNets; CIDR and subnet design; public and private subnets; IP planning; route tables; internet and NAT gateways; interfaces; AZ-aware design.
- **Module 4 — Routing & Connectivity:** Cloud routing; internet connectivity; site-to-site VPN; private connectivity; peering and transit; hybrid; DNS and private zones; load balancing.
- **Module 5 — Cloud Network Security:** Security groups; network ACLs; IAM; least privilege; encryption in transit; private service access; bastions; audit logs; threat modelling.
- **Module 6 — Compute & Deployment:** VMs; containers; intro Kubernetes networking; autoscaling; load balancers; high availability; deployment workflows.
- **Module 7 — Monitoring & Troubleshooting:** Native monitoring; flow logs; metrics and alerts; DNS and connectivity debugging; routing failures; firewall analysis; resource health.
- **Module 8 — Automation & IaC:** IaC concepts; Terraform intro; reusable configuration; version control; environment separation; safe validation; secrets management.
- **Module 9 — Cost Optimization & Reliability:** Tagging; budget alerts; cost-aware architecture; resilience; backup and DR; availability planning.
- **Module 10 — Cloud Projects:** Simple VNet; segmented architecture; secure public/private subnets; simulated hybrid link; security review; architecture, cost and risk documentation.

Course 3 — Ethical Hacking & Penetration Testing (/courses/ethical-hacking)

Category Cybersecurity · Intermediate · 11 modules (one optional) · 89 lessons · ≈59 h. A visible rules-of-engagement banner: exercises only on systems the learner owns or is explicitly authorized to test, using isolated labs and synthetic data.

- **Module 1 — Cybersecurity Fundamentals:** CIA; threats, vulnerabilities, risk; controls; authN/authZ; MFA and passwords; malware and social engineering awareness; policies and incident response; ethics and rules of engagement.
- **Module 2 — Networking for Security:** OSI/TCP-IP review; addressing; TCP, UDP, ports; DNS and HTTP/HTTPS; firewalls, NAT, VPNs; packet analysis; segmentation; secure architecture.

- **Module 3 — Linux & Command Line:** Distributions; navigation; permissions and ownership; users and groups; processes and services; packages; shell; logs; safe scripting.
- **Module 4 — Security Lab Setup:** VMs; isolated networks; snapshots; intentionally vulnerable targets; safe test data; lab documentation; authorization and scope.
- **Module 5 — Reconnaissance & Assessment:** Scope; passive information gathering; public information; asset inventories; exposure assessment in authorized labs; threat modelling; evidence; risk prioritisation.
- **Module 6 — Vulnerability Assessment:** Vulnerability and misconfiguration concepts; scanners in controlled labs; false positives; CVE and CVSS; patching; configuration review; verification; remediation.
- **Module 7 — Web Application Security:** HTTP structure; cookies and sessions; auth and access-control weaknesses; input validation; injection; XSS; CSRF; secure headers and TLS; OWASP Top 10; secure coding.
- **Module 8 — System & Network Hardening:** Baselines; service minimisation; patching; least privilege; SSH hardening; firewall policies; logging and alerting; backup; defensive monitoring.
- **Module 9 — Penetration Testing Methodology:** Pre-engagement and written authorization; scope and RoE; discovery; controlled validation; evidence handling; risk classification; reporting; retest and closure.
- **Module 10 — Reporting & Capstone:** Executive summaries; technical findings; reproduction steps in authorized labs; impact; evidence without sensitive data; remediation; retest; final report.
- **Module 11 — Optional: Beyond the Core:** Digital forensics; SOC and SIEM; incident response; cloud security; secure software development; certifications and career pathways.

Do not promise certification, employment or professional qualification. Instructor information only when supplied.

9. Course detail page template

- Hero: breadcrumbs, level and category chips, title, overview, “Start the first lesson / Continue learning” and “Enroll for free / Enrolled” buttons, personal progress bar when enrolled, illustration card with calculated counts and estimate note.
- Rules-of-engagement banner (security course).
- Learning objectives; Who it's for; Prerequisites; Skills you'll practise (chips).
- Full curriculum accordion: numbered modules with project/optional chips, summary, key terms, numbered lesson links, “Full lesson” badges, completion ticks; Expand all / Collapse all; deep link ?m=n opens and scrolls to a module.
- Practical exercises & projects; Tools & software; Course FAQ; instructor note; next-course CTA (or contact CTA); related courses.

10. Lesson player (/learn/:course/:lesson)

- Left sidebar with the course outline (current module expanded, completion ticks), hidden on phones; breadcrumbs; module and lesson number; estimated minutes; Bookmark toggle.
- Fully authored lessons follow nine parts: 1 What it means, 2 Why it matters, 3 Explanation, 4 Practical example, 5 Diagram (custom SVG), 6 Guided exercise, 7 Knowledge check (quiz with explanations), 8 Common mistakes, 9 Summary and further reading.
- Authored lessons: What is a computer network?; The seven OSI layers; Subnet masks and subnet calculations; What cloud computing is; Shared responsibility model; Virtual private clouds; Confidentiality, integrity and availability; Ethics and rules of engagement; Injection vulnerabilities (defensive, parameterised queries).
- Other lessons are clearly labelled **outline lessons**: where it fits, lessons in the module, key terms, linked lab, module check quiz.
- Mark complete (auto-advances, auto-enrolls), previous/next cards, quiz results saved, “last lesson” saved for Continue learning; sign-in prompt when previewing.

11. Interactive learning lab (/labs)

Ten exercises with real logic, validation, feedback, reset controls and mobile layouts. Tabs on desktop, a dropdown selector on phones; deep link via ?t=id. Every simulated exercise carries the note: “Simulated with sample data in your browser. No real network, DNS, cloud or scan activity takes place.”

#	Exercise	Required behaviour
1	Subnetting calculator	IPv4 + CIDR (slider and mask dropdown): network, broadcast, mask, wildcard, first/last host, usable hosts (handles /31, /32), total, address type (private, public, loopback, APIPA, CGNAT, multicast), colour-coded binary view; random practice questions with worked answers.
2	OSI model explorer	Seven clickable layers showing PDU, devices, protocols, real-world example and typical problems; 10-question “Which layer?” quiz.
3	Packet journey	Two scenarios (load a web page, ping the gateway); step / back / auto-play / reset; animated packet; explanations of ARP, MAC rewrite, NAT, TTL, longest-prefix match. Vertical diagram on phones.
4	DNS lookup simulator	Sample zones under .example and .test; A, AAAA, MX, CNAME, TXT; animated root → TLD → authoritative steps, CNAME following, NXDOMAIN/NODATA, 300 s resolver cache with clear-cache.
5	Linux command quiz	Simulated shell with virtual filesystem (pwd, ls -la, cd, cat, grep -i, chmod numeric, mkdir, touch, whoami, id, ip a, ss, ps, history, clear); six tasks auto-detected; history with arrow keys; sudo disabled.
6	Cloud architecture builder	Place IGW, NAT, load balancer, two app servers, database and bastion in public/private subnets across two AZs; choose DB and SSH rules; eight live checks with explanations; “show a passing design”; cost note.
7	Firewall rule evaluation	Six ordered rules with CIDR matching and first-match semantics; seven synthetic packets; predict allow/deny, then check with matched rule; reorder rules to see effects.
8	Knowledge quizzes	Ten random questions per course from module checks and lesson quizzes; results saved to the dashboard when signed in.
9	Vulnerability prioritisation	Six sample findings (CVSS, exposure, asset, exploitability); reorder; compare with a context-weighted reference ranking and explanation.
10	Security report writing	Fictional authorized lab scenario; fields for title, severity, asset, CVSS, description, evidence, impact, remediation; nine-point review (incl. no credentials in evidence, parameterised-query fix) and a printable-style finding preview; example answer.

12. Other public pages

- **About:** what MND CODE does, how it teaches (clarity, practice, responsibility, honest claims), company facts (MND CODE Pvt Ltd, established 01 April 2023, MNDCODE.in, tagline).
- **All courses:** filters for search text, category, difficulty, estimated duration; live result count; empty state with reset.
- **Learning paths:** Network Administrator, Cloud Network Engineer, Security Analyst & Pentester — audience, five ordered steps, linked courses, total estimated hours.
- **Projects:** five educational briefs (small office network, VLAN and routing simulation, secure cloud architecture, Linux hardening checklist, authorized web assessment lab) with level, course and deliverables.
- **Resources:** filterable accordions — common ports, IPv4 subnet reference (generated), Linux commands, bottom-up troubleshooting, reading CVSS, avoiding cloud charges.
- **Blog:** paginated list (3 per page) and article pages with original content: why subnetting matters in the cloud, building a first isolated security lab, reading a routing table, severity vs risk.

- **FAQ:** ten questions incl. experience, order, equipment, are labs real, certification, legality, cloud charges, how progress is saved, duration, contact.
- **Contact:** name, email, optional phone, course interest, subject, message (2000-char counter); inline validation on blur; loading state; success/failure messages; privacy notice; contact details panel showing “To be confirmed” until configured. Never invent emails, phones, addresses or social profiles.
- **Privacy and Terms:** template policies (data collected, use, retention, rights incl. India's DPDP Act 2023, security, demo notice; acceptable use for security training referencing the IT Act 2000, no guarantees, third-party charges), marked for legal review.

13. Accounts and student dashboard

- **Register:** name, email (duplicate check), password (8+ chars, upper, lower, digit), confirm, terms checkbox including the authorized-testing pledge.
- **Sign in:** demo-account quick fill, generic error message, 5-attempt lockout for 30 s, redirect to ?next.
- **Dashboard:** welcome, four stats (enrolled, lessons done, average quiz, streak), Continue learning card, enrolled courses with progress, recent activity, 28-day activity grid, bookmarks; sample activity labelled.
- **My courses:** every course with enrolled state, progress, Start/Continue, curriculum link, enroll and unenroll (with confirmation; progress kept).
- **Progress & achievements:** six achievements computed from real activity (first lesson, ten lessons, perfect quiz, 3-day streak, module complete, all three courses), per-module progress bars, quiz-attempt table; certificates explicitly not issued until a real mechanism exists.
- **Settings:** edit profile, change password (verifies current), reset progress, delete account, sign out — destructive actions confirmed in a dialog.
- Streaks are calculated from actual activity days only.

14. Admin dashboard (role: admin)

- Banner explaining prototype storage vs production server-side checks; sidebar with unread-message badge.
- **Overview:** stats (courses, published lessons, students, unread messages); announcement editor (on/off, label, text, internal link validated, link text); homepage SEO title and description with character guidance; audit log of sensitive actions.
- **Courses:** table with counts and question totals; create (title, category, level, description, modules entered as “Module: lesson; lesson”), edit, delete/hide with confirmation, restore defaults.
- **Lessons:** per-course accordion; publish/unpublish (drafts hidden from students), view, add lessons with draft/published status, delete custom lessons.
- **Students:** searchable table; enroll a student in a course; remove account (confirmation; cannot remove yourself).
- **Messages:** All / Unread / Read filters; expand marks as read; toggle read state; copy email; delete with confirmation; empty state.

15. Mobile and responsive requirements

Design mobile-first and test at 375, 390, 768, 1024 and 1366 px. The page body must never scroll sideways; side gutters ≥16–20 px including safe-area insets.

Breakpoint	Behaviour
≤ 1080 px	Desktop nav replaced by hamburger + full-screen mobile menu; Sign In moves into the menu.
≤ 960 px	4-column and 3-column grids become 2 columns; process band 2 columns; footer 2 columns.
≤ 900 px	Hero, course hero and contact layouts stack to one column; dashboard sidebars become a horizontal scrolling pill row; lesson sidebar hidden; lab panes stack.

Breakpoint	Behaviour
≤ 860 px	Roadmap becomes a vertical timeline; bento grid single column.
≤ 640 px (phones)	Compact two-line announcement; 60 px header without logo sub-line; full-width hero and CTA buttons; shorter hero visual with smaller telemetry; highlight, lab-preview and project rows become **swipeable snap carousels**; stats shown two per row; **tables convert to labelled stacked cards**; lab tabs become a dropdown; packet diagram turns vertical; tighter section, card, accordion and dialog padding; 40 px minimum touch targets; form fields at 16 px so iOS does not zoom.
≤ 380 px	Result grids single column; stat numbers reduced.

- Wide content (diagrams, code) scrolls inside its own container only.
- Sticky header respects the safe-area top inset; fixed buttons respect bottom/side insets.
- Images, SVGs and aspect-ratio boxes max-width 100%; flex/grid children holding text get min-width 0.

16. Accessibility, SEO and performance

- WCAG 2.2 AA: semantic landmarks, skip link, visible cyan focus ring, keyboard-operable dropdown, dialogs, tabs (arrow keys) and palette; labels, aria-invalid and inline errors; aria-current in navigation and breadcrumbs; live regions for results; descriptive alt/aria labels for visuals.
- Unique page titles and meta descriptions per route; Open Graph tags; canonical URLs, sitemap.xml and robots.txt in production (static rendering or SSR, since hash routes are not indexed separately); structured data only where genuinely applicable.
- Fast and light: inline assets, Google Fonts with display=swap, canvas animation paused when reduced motion is on, route-level rendering, error boundaries; analytics only if configured with privacy controls.

17. Content integrity rules

- Do not invent certifications, accreditations, partnerships, placement statistics, ratings, learner counts, testimonials, instructors or registration claims.
- Label sample projects, sample activity, demo accounts and simulated lab results. Never imply a simulated result came from a real scan, DNS query or cloud deployment.
- Write original, technically accurate, beginner-friendly content; no filler paragraphs.
- All security exercises restricted to owned or explicitly authorized systems, isolated labs and synthetic data.

18. Production backend

- PostgreSQL tables: users, sessions, courses, modules, lessons, quiz_questions, enrollments, lesson_progress, quiz_attempts, bookmarks, contact_submissions (with retention date), announcements, audit_events.
- REST endpoints: auth (register, login, logout, me), courses and lessons, enrollments, progress, quiz attempts, bookmarks, contact (rate-limited), /api/admin/* guarded by role and audited.
- Security: Argon2id hashing, HTTP-only Secure SameSite cookies, CSRF protection, rate limiting, parameterised queries, input validation and output encoding, secret manager, migrations, backups, structured logging, privacy and retention controls.
- Provide .env.example without real secrets.

19. Deliverables and acceptance checklist

- Complete responsive website (single-file index.html for the prototype) with all routes above.
- README: local run, configuration (SITE object), static deployment (Netlify/Vercel/Cloudflare Pages/Nginx), MNDCODE.in DNS + HTTPS + www redirect + email records, backend plan.
- .env.example and schema.sql.

- Tested checklist: every route renders with a unique title and no console errors; no horizontal scroll at 375 px; all forms validate with clear messages; search, filters, accordions, enrollment, progress, bookmarks, quizzes and all ten lab exercises work; admin create/edit/delete flows use confirmations and are logged; reduced-motion and keyboard navigation verified.

Final instruction: start with the design system and homepage, then the shared layout and all routes; build and verify each course page, the lab, the student dashboard and the admin workflow; test on mobile, tablet and desktop. Do not stop at a hero section, do not ship non-working buttons, and do not invent business credentials or results. The finished experience should make MND CODE look like a distinctive, modern, trustworthy technical learning brand.